

Lesson 18 — If / Then / Else

Matches LEGO Teacher Portal lesson: "Sailing to Safety".

Objective (I can...): I can use if/then/else so my model does one thing OR another. **CS concept:** Two-way branching (else) · **Builds on:** Lesson 17 (true/false).

AI Connection (2 min): AI branches: IF it recognizes you, THEN unlock; ELSE stay locked. *Impacts prompt:* "Face-unlock helps you get in fast — who might it help most, and could it bother or lock out anyone?"

Materials: LEGO CS & AI kit + device per pair, trigger object/color card. **Model reuse:** Conditionals model from Lesson 16 — code only.

60-minute flow - Warm-up (8): "Two doors" — conditions with two outcomes ("if green, go; else stop"); pairs act out both branches. - **Concept + demo (10):** Demo if/then/else: object near → green light; else → red light. Run both branches. - **Build/Code (30):** Pairs code an if/then/else with two clearly different outcomes from one condition. *Swap at 15 min.* - **Challenge (7):** Make each branch a two-step action (light + sound). - **Share + pack-up (5):** Pairs demo both branches and say which condition chose which. Model stays built.

Pair roles: Driver codes the "then," Navigator plans the "else"; swap at 15 min so each owns a branch. *(Trio: third owns testing both.)*

Differentiation: *Support* — the "then" filled in; students add only the "else," framed as pseudocode. *Extension / fast-finisher* — nest an if inside a branch. *Reteach* — trigger each branch by hand and label it before coding.

Exit check: Pair's model shows two different outcomes correctly chosen by one condition. **At-home (optional):** Write an if/then/else rule for choosing what to wear. ~10 min.
